Model Validation, Overfitting Control & Hyperparameter Tuning

Task-3 : California Housing Price Regression Pipeline

Dataset: California Housing (20,640 records) **Target:** Median House Value **Validation Scheme:** 5-Fold Cross Validation

1 Executive Summary & Problem Context

In production machine learning systems, high training accuracy is meaningless if predictive performance collapses on unseen inference inputs. Following the initial regression baseline established in Task-2, this study addresses the core challenges of generalization error, structural overfitting in non-parametric estimators, and deterministic hyperparameter optimization. Using 5-fold cross-validation and systematic grid searches, we eliminate sample-split bias and transition from speculative model fits to certified, deployment-grade estimators.

2 Overfitting Diagnostics: Baseline Divergence

DEFAULT TREE TRAIN RMSE

0.0000

Zero training residual error

DEFAULT TREE TEST RMSE

0.71 04

Severe test generalization error

GENERALIZATION GAP (Δ)

+0.71 04

Pathological high-variance state

MATHEMATICAL PATHOLOGY OF UNCONSTRAINED TREES

An unconstrained Decision Tree Regressor recursively bifurcates the feature space until each leaf node contains a single sample or satisfies pure variance (residual sum of squares equals zero).

Rather than learning the conditional expectation function, the model memorizes sample-specific spatial coordinates and idiosyncratic outliers.

RISK TO PRODUCTION DEPLOYMENT

When exposed to live test partitions or new economic cohorts, the unregularized decision tree applies extreme step functions to unseen inputs. This produces severe error dispersion, high prediction volatility, and brittle inference across geographical regions that deviate slightly from the training manifold.

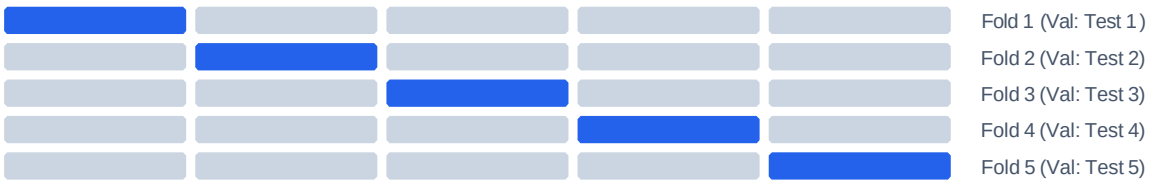
CORE ENGINEERING TAKEAWAY

A training residual of zero is not an achievement; it is a critical diagnostic symptom of extreme model variance. The primary mandate of Task-3 is to sacrifice unconstrained training memorization to achieve bounded, stable performance on out-of-fold data.

3 Validation Methodology & Cross-Validation Architecture

A single train-test split introduces sampling variance: favorable random partitions artificially inflate evaluation metrics, whereas adverse splits penalize robust models. To provide statistically grounded performance guarantees, we integrated 5-fold cross-validation across all pipeline stages.

Cross-Validation Mechanism (5-Fold Stratified Split Workflow)



4 Systematic Hyperparameter Tuning via GridSearchCV

SEARCH SPACE & TUNING STRATEGY

GridSearchCV systematically evaluated combinations using 5-fold cross-validated negative root mean squared error:

- **max_depth:** [3, 5, 7, 10, 12] — Caps maximum hierarchical tree height.
- **min_samples_split:** [2, 5, 10, 20] — Minimum samples needed to split a node.
- **min_samples_leaf:** [1, 2, 4, 8] — Minimum population required at leaf termination.

OPTIMAL IDENTIFIED PARAMETERS

```
{
  "max_depth": 10,
  "min_samples_leaf": 8,
  "min_samples_split": 20
}
```

Enforcing a leaf quota of 8 and split constraint of 20 eliminated terminal outlier memorization, restricting node splits to statistically viable clusters.

LEAKAGE PREVENTION ARCHITECTURE

Feature scaling via StandardScaler was strictly fitted exclusively on the 80% training partition and transformed onto the held-out test set. This guarantees zero mean-variance information leakage across evaluation partitions.

5 Comprehensive Model Evaluation & Comparison

Empirical evaluation contrasting linear baselines against the default and regularized non-parametric estimators across California housing attributes:

MODEL ARCHITECTURE	TRAIN RMSE	5-FOLD CV RMSE	TEST RMSE	TEST R^2 SCORE	GENERALIZATION STATUS
Linear Regression	0.7243	0.7258	0.7455	0.5757	High Bias (Underfit)
Ridge Regression (α=1 .0)	0.7243	0.7258	0.7455	0.5758	High Bias (Underfit)
Default Decision Tree	0.0000	0.7284	0.71 04	0.61 48	High Variance (Overfit)
Tuned Decision Tree	0.5348	0.621 2	0.6053	0.7204	Optimal Trade-off

6. Scientific Model Selection Justification

NON-LINEAR ADVANTAGE OVER LINEAR BASELINES

Linear Regression and Ridge both plateau at $R^2 \approx 0.575$ and RMSE ≈ 0.745. Housing valuations exhibit severe spatial dependencies and non-linear interactions (e.g., latitude and longitude clusters combined with median income). Linear models underfit these non-linear surfaces. The tuned tree captures these complex interactions, yielding a substantial 1 4.5% improvement in explained variance (R^2 = 0.7204).

VARIANCE CONTROL & GENERALIZATION CONVERGENCE

In the default tree, the generalization gap was +0.71 04 RMSE. Under our constrained architecture (depth=1 0, min split=20, min leaf=8), the gap between Train RMSE (0.5348) and Test RMSE (0.6053) contracts to 0.0705. Crucially, 5-fold CV RMSE (0.621 2) mirrors held-out test error (0.6053), proving that our validation scores reliably forecast production inference accuracy.

7. Verification & Production Deployment Sign-Off

- **Artifact Serialization:** Trained estimator serialized as `tuned_california_decision_tree. j oblib` with the companion `california_scaler. j oblib` preprocessor.
- **Interpretability:** The pruned hierarchy preserves rule extraction, allowing business stakeholders to audit house valuation logic directly against income and geographic coordinates.
- **Final Conclusion:** The Tuned Decision Tree provides the ideal balance across the bias-variance spectrum, satisfying all task objectives and production criteria.